Concavity, Curve Sketching, and Optimization

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Increasing/decreasing and the first derivative test**

For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. $f' > 0$ on $(-\infty, 1)$, $f' < 0$ on $(1, 3)$, $f' > 0$ on $(3, \infty)$. So f is increasing on $(-\infty, 1) \cup (3, \infty)$, decreasing on $(1, 3)$. $x = 1$ is a local max (sign $+ \to -$); $x = 3$ is a local min (sign $- \to +$).

2 **[Foundational]** • **Target: Concavity and the second derivative test**

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f'(x) = 4x^3 - 12x^2$; $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ on $(-\infty, 0)$, $f'' < 0$ on $(0, 2)$, $f'' > 0$ on $(2, \infty)$. Concave up on $(-\infty, 0) \cup (2, \infty)$, concave down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$ (both sign changes).

3 **[Foundational]** • **Target: Curve sketching**

For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

$f'(x) = 3x^2 - 6x = 3x(x-2)$: critical points $x = 0, 2$. $f''(x) = 6x - 6$: zero at $x = 1$. Sign chart: $f' > 0$ on $(-\infty, 0)$, $f' < 0$ on $(0, 2)$, $f' > 0$ on $(2, \infty)$ — local max at $x = 0$ ($f(0) = 0$), local min at $x = 2$ ($f(2) = -4$). $f'' < 0$ on $(-\infty, 1)$ (concave down), $f'' > 0$ on $(1, \infty)$ (concave up) — inflection point at $x = 1$ ($f(1) = -2$).

4 **[Foundational]** • **Target: Optimization problems**

A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

Let x = side parallel to river, y = each perpendicular side. Constraint: $x + 2y = 200$, so $x = 200 - 2y$. Area $A = xy = (200 - 2y)y = 200y - 2y^2$. $A'(y) = 200 - 4y = 0 \Rightarrow y = 50$. $x = 200 - 100 = 100$. Maximum area occurs at $x = 100$ m, $y = 50$ m (verified max since $A'' = -4 < 0$).

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5 [Foundational] • Target: Optimization problems

An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.

$V(x) = x(12-2x)^2$ for $0 < x < 6$. $V'(x) = (12-2x)^2 + x \cdot 2(12-2x)(-2) = (12-2x)[(12-2x) - 4x] = (12-2x)(12-6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. $V(2) = 2(8)^2 = 128$ in^3, the maximum.

6 [Foundational] • Target: Optimization: minimizing cost or distance

Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

$D(x) = x^2 + (2x-3)^2 = x^2 + 4x^2 - 12x + 9 = 5x^2 - 12x + 9$. $D'(x) = 10x - 12 = 0 \Rightarrow x = \frac{6}{5}$.
$y = 2\left(\frac{6}{5}\right) - 3 = \frac{12}{5} - 3 = -\frac{3}{5}$. Closest point: $\left(\frac{6}{5}, -\frac{3}{5}\right)$ (confirmed min since $D'' = 10 > 0$).

7 [Foundational] • Target: Optimization: minimizing cost or distance

A closed cylindrical can must have volume 16π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.

From V: $h = \frac{16}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{16}{r^2} = 2\pi r^2 + \frac{32\pi}{r}$. $S'(r) = 4\pi r - \frac{32\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{32\pi}{r^2} \Rightarrow r^3 = 8 \Rightarrow r = 2$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Points of inflection and graph behavior

Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

$f'(x) = 3x^2 \geq 0$ everywhere, so f is never decreasing — $x = 0$ is not a local extremum. However $f''(x) = 6x$ changes sign at $x = 0$ (from concave down to concave up), so $x = 0$ IS an inflection point. This shows concavity change and extrema are independent features.

9 [Intermediate] • Target: Visualizing extrema and concavity together

The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.

The graph rises to a peak at $x = 0$ ($f = 0$), then falls, changing its bend from concave-down to concave-up right at $x = 1$ ($f = -2$, the inflection point), before reaching a valley at $x = 2$ ($f = -4$) and rising again — visually confirming all three features found algebraically.

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10 **[Intermediate]** • **Target: A capstone optimization problem**

A rectangular box with a square base and no top must have volume 500 cm^3. Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.

$S(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}$. $S'(x) = 2x - \frac{2000}{x^2} = 0 \Rightarrow 2x^3 = 2000 \Rightarrow x^3 = 1000 \Rightarrow x = 10$ cm.

11 **[Intermediate]** • **Target: Second derivative test failures**

For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

$f'(x) = 4x^3 = 0$ gives $x = 0$. $f''(x) = 12x^2$; $f''(0) = 0$, so the Second Derivative Test gives no information. Using the First Derivative Test: $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$, so $x = 0$ is a local (and in fact absolute) minimum.

12 **[Intermediate]** • **Target: A capstone curve-sketching problem**

For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x-2)(x+1)$: critical points $x = -1, 0, 2$. $f''(x) = 36x^2 - 24x - 24$.
$f''(-1) = 36 + 24 - 24 = 36 > 0$: local min. $f''(0) = -24 < 0$: local max.
$f''(2) = 144 - 48 - 24 = 72 > 0$: local min.

13 **[Intermediate]** • **Target: Curve sketching**

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0) = 2$.

Identify $P(x) = -1$ and $Q(x) = e^{3x}$. Compute integrating factor: $\mu(x) = e^{\int -1\,dx} = e^{-x}$. Multiply through: $\frac{d}{dx}[e^{-x}y] = e^{-x}e^{3x} = e^{2x}$ Integrate both sides: $e^{-x}y = \int e^{2x}\,dx = \frac{1}{2}e^{2x} + C \Rightarrow y(x) = \frac{1}{2}e^{3x} + Ce^{x}$ Apply initial condition $y(0) = 2$: $2 = \frac{1}{2}(1) + C(1) \Rightarrow C = \frac{3}{2}$ Thus, $y(x) = \frac{1}{2}e^{3x} + \frac{3}{2}e^{x}$.

14 **[Intermediate]** • **Target: Optimization problems**

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

Set up the ratio: $\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n}\right| = \lim_{n\to\infty} \left(\frac{4^{n+1}}{4^n} \cdot \frac{n!}{(n+1)n!}\right) = \lim_{n\to\infty} \frac{4}{n+1} = 0.$ Since \(\rho = 0 Ratio Test, the series converges absolutely.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

The field is conservative with potential function $f(x, y, z) = x^3y + 2z$. By the Fundamental Theorem for Line Integrals: $\int_C abla f \cdot d\mathbf{r} = f(2, 1, 3) - f(0, 0, 1)$ Evaluate at endpoints: $f(2, 1, 3) = 2^3(1) + 2(3) = 8 + 6 = 14$ $f(0, 0, 1) = 0^3(0) + 2(1) = 2$ Therefore: $\int_C \mathbf{F} \cdot d\mathbf{r} = 14 - 2 = 12$

16 **[Challenging] • Target: Points of inflection and graph behavior • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

Compute the limit of the general term as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos(0) = 1$ Because $\lim_{n\to\infty} a_n = 1e0$, the series diverges by the Divergence Test.

17 **[Challenging] • Target: Visualizing extrema and concavity together • Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

Set $r = 0$: $3\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$. By the Tangents at the Pole Theorem, when $r(\theta_0) = 0$ and $r'(\theta_0)eq0$, the line $\theta = \theta_0$ is the tangent line. Thus, the tangent line at the pole is $\boldsymbol{\theta = \frac{\pi}{2}}$ (the vertical line $x = 0$).

18 **[Challenging] • Target: A capstone optimization problem • Cross-Lesson: 8.01 Area Between Curves**

Find the perpendicular distance from point $P(3, 1, -2)$ to plane $\pi : 2x - 2y + z = 5$.

Apply the point-to-plane distance formula: $D = \frac{|aP_x + bP_y + cP_z - d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|2(3) - 2(1) + 1(-2) - 5|}{\sqrt{2^2 + (-2)^2 + 1^2}} = \frac{|6 - 2 - 2 - 5|}{\sqrt{4 + 4 + 1}} = \frac{|-3|}{3} = 1.$

19 **[Challenging] • Target: Second derivative test failures • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Level 1 · Foundations Find $\frac{dy}{dx}$ in terms of t for the parametric curve $x = 2\cos t$, $y = 5\sin t$ for $t \in (0, \pi)$.

Compute derivatives with respect to t: $\frac{dx}{dt} = -2\sin t$ $\frac{dy}{dt} = 5\cos t$ By Theorem 9.1: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{5\cos t}{-2\sin t} = -\frac{5}{2}\cot t$

20 **[Challenging] • Target: A capstone curve-sketching problem • Cross-Lesson: 1.01 Limits Graphically and Numerically**

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Here $f(x) = 2x$ and $f'(x) = 2$. The arc length element is $ds = \sqrt{1 + (f'(x))^2}\, dx = \sqrt{1 + 2^2}\, dx = \sqrt{5}\, dx$. The surface area formula gives: $S = 2\pi\int_0^3 f(x)\sqrt{1 + [f'(x)]^2}\, dx = 2\pi\int_0^3 (2x)\sqrt{5}\, dx = 4\sqrt{5}\pi\left[\frac{x^2}{2}\right]_0^3 = 4\sqrt{5}\pi\left(\frac{9}{2}\right) = 18\sqrt{5}\pi$ Verification via cone geometry: The shape is a right circular cone with radius $r = f(3) = 6$ and slant height $l = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$. Lateral are.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Increasing/decreasing and the first derivative test

Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.

a. a local maximum

b. cannot be determined without more information

c. **a local minimum ✓**

d. an inflection point

$f''(x) = 6x + 12$, so $f''(-1) = -6 + 12 = 6$, which is positive: $x = -1$ is a local minimum.

22 [Foundational] • Target: Concavity and the second derivative test

Find the nonzero inflection point of $f(x) = x^4 + 6x^3$.

a. -3 ✓

b. -6

c. -2

d. 6

$f''(x) = 12x^2 + 36x = 6x(2x + 6)$. Zero at $x = 0$ and $x = -3$; the nonzero inflection point is $x = -3$.

23 [Foundational] • Target: Curve sketching

A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 26

b. 100

c. 25 ✓

d. 50

$A(y) = (100 - 2y)y = 100y - 2y^2$. $A'(y) = 100 - 4y = 0 \Rightarrow y = \frac{100}{4} = 25$ m.

24 [Foundational] • Target: Optimization problems

Find the x-coordinate of the point on the line $y = -5x - 4$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{4}{5}$

b. $\frac{2}{13}$

c. $-\frac{10}{13}$ ✓

d. $\frac{10}{13}$

$D(x) = x^2 + (-5x - 4)^2 = (26)x^2 + 40x + 16$. $D'(x) = 52x + 40 = 0 \Rightarrow x = \frac{-20}{26} = -\frac{10}{13}$.

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25 [Foundational] • Target: Optimization: minimizing cost or distance

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.

a. **2** ✓

b. 3

c. 4

d. 5

To find the x-intercepts, we set $f(x) = 0$. Thus, we solve the quadratic equation $x^2 - 4x + 3 = 0$. This can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-coordinates of points A and B. Therefore, the points are (1, 0) and (3, 0). The distance between these two points on the x-axis is the absolute difference of their x-coordinates: $|3 - 1| = 2$.

26 [Foundational] • Target: Points of inflection and graph behavior

Local min at $x = c$ by First Derivative Test requires f':

a. positive then negative

b. **negative then positive** ✓

c. positive everywhere

d. zero everywhere

Applying Applications of Derivatives principles, Sign change from $-$ to $+$. Therefore, the verified correct option is negative then positive.

27 [Foundational] • Target: Visualizing extrema and concavity together

$f(x, y) = xy$: at origin, the second derivative test gives $D = f_{xx}f_{yy} - f_{xy}^2 = 0 - 1 = -1 < 0 \Rightarrow$

a. **saddle point** ✓

b. local max

c. local min

d. inconclusive

Applying Partial Derivatives and Gradient principles, $D < 0 \Rightarrow$ saddle. Therefore, the verified correct option is saddle point.

28 [Foundational] • Target: A capstone optimization problem

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. **inconclusive** ✓

When $f''(c) = 0$ too, the test gives no information; another method is needed.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Second derivative test failures**

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

c. zero to zero

d. **positive to negative ✓**

f increasing then decreasing through c means c is a local maximum.

30 **[Intermediate]** • **Target: A capstone curve-sketching problem**

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:

a. inflection point

b. no conclusion

c. **local maximum ✓**

d. local minimum

Concave down ($f''(c) < 0$) at a critical point means a local maximum.

31 **[Intermediate]** • **Target: Increasing/decreasing and the first derivative test**

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:

a. **inconclusive ✓**

b. proves local min

c. proves local max

d. proves inflection

Test fails; need higher derivatives or sign analysis.

32 **[Intermediate]** • **Target: Concavity and the second derivative test**

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:

a. local max

b. **local min ✓**

c. inflection point

d. saddle point

$f'' > 0$ at critical point $\Rightarrow$ local min.

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33 [Intermediate] • Target: Curve sketching

The second derivative test fails ($D = 0$). This means the critical point is:

a. always a saddle

b. **indeterminate without further analysis ✓**

c. always min

d. always max

$D = 0$: inconclusive — need higher-order analysis.

34 [Intermediate] • Target: Optimization problems

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. **local min ✓**

b. local max

c. inflection

d. saddle

$f'' > 0$: concave up $\Rightarrow$ local min.

35 [Intermediate] • Target: Optimization: minimizing cost or distance

The second derivative test: $D = f_{xx}f_{yy} - f_{xy}^2$. If $D > 0$ and $f_{xx} > 0$, the point is a:

a. saddle point

b. **local min ✓**

c. local max

d. inflection point

$D > 0, f_{xx} > 0 \Rightarrow$ local minimum.

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36 [Intermediate] • Target: Points of inflection and graph behavior

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

a. **4** ✓

b. 2

c. 3

d. 6

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x -coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

37 [Intermediate] • Target: Visualizing extrema and concavity together

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.

a. 1

b. -2

c. **2** ✓

d. -1

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $f(x) = 2x^2 - 8x + k$, so $a = 2$ and $b = -8$. The x-coordinate of the vertex is given as 2, which is consistent: $x = -\frac{-8}{2(2)} = \frac{8}{4} = 2$. The y-coordinate of the vertex is $f(2)$. We are given that the y-coordinate of the vertex is -6. Therefore, $f(2) = -6$. Substitute $x = 2$ into the function: $f(2) = 2(2)^2 - 8(2) + k$. $f(2) = 2(4) - 16 + k$ $f(2) = 8 - 16 + k$ $f(2) = -8 + k$. Since $f(2) = -6$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A capstone optimization problem • Cross-Lesson: 8.01 Area Between Curves

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the x-coordinate of the vertex of the parabola.

a. 1

b. **2** ✓

c. 3

d. 4

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, the x-coordinate of the vertex is the midpoint of the roots. The roots are the x-intercepts, which can be found by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring this gives $(x - 1)(x - 3) = 0$, so the roots are $x = 1$ and $x = 3$. The midpoint is.

39 [Challenging] • Target: Second derivative test failures • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?

a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$ ✓

d. $(3, 4)$

The graph of $f(x)$ passes through the point $(1, 0)$. To verify this, we can observe the graph and identify the coordinates where the curve intersects the x-axis at $x = 1$. Let's check the other options. The point $(-2, 3)$ is not on the curve as the function value at $x = -2$ is approximately 5. The point $(0, -1)$ is not on the curve as the function value at $x = 0$ is approximately 3. The point $(3, 4)$ is not on the curve as the function value at $x = 3$ is approximately 0. Therefore, only $(1, 0)$ lies on the.

40 [Challenging] • Target: A capstone curve-sketching problem • Cross-Lesson: 1.01 Limits Graphically and Numerically

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.

a. **2 units** ✓

b. 3 units

c. 4 units

d. 6 units

To find the points where the graph intersects the x-axis, we set $f(x) = 0$. $x^2 - 6x + 8 = 0$. Factoring the quadratic equation, we get $(x - 2)(x - 4) = 0$. Thus, the x-intercepts are $x = 2$ and $x = 4$. Let these points be A and B. The distance between A and B is $|4 - 2| = 2$ units. To find the y-intercept, we set $x = 0$. $f(0) = 0^2 - 6(0) + 8 = 8$. So, point C is (0, 8). The question asks for the length of the line segment AB, which is the distance between the x-intercepts.

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41 [Challenging] • Target: Increasing/decreasing and the first derivative test • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

The graph of the function $g(x) = x^2 - 6x + k$ is shown below. The graph intersects the x-axis at points P and Q. Point P has coordinates $(2, 0)$. Find the value of the parameter k.

a. 1

b. **2** ✓

c. 4

d. 8

The graph of the quadratic function $g(x) = x^2 - 6x + k$ intersects the x-axis at points where $g(x) = 0$. We are given that one of these intersection points is P, with coordinates $(2, 0)$. This means that when $x = 2$, $g(x) = 0$. We can substitute these values into the function to find the value of k: $g(2) = (2)^2 - 6(2) + k$ $0 = 4 - 12 + k$ $0 = -8 + k$ $k = 8$ Therefore, the value of the parameter k is 8.

42 [Challenging] • Target: Concavity and the second derivative test • Cross-Lesson: 2.01 The Definition of the Derivative

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?

a. 10 ✓

b. 12

c. 14

d. 16

The area of the region is given by the definite integral of $f(x)$ from $x = 1$ to $x = 2$. Area $= \int_1^2 (3x^2 + 1)\,dx$ First, find the antiderivative of $3x^2 + 1$: Antiderivative $= \frac{3x^3}{3} + x = x^3 + x$ Now, evaluate the definite integral using the Fundamental Theorem of Calculus: Area $= [x^3 + x]_1^2$ Area $= (2^3 + 2) - (1^3 + 1)$ Area $= (8 + 2) - (1 + 1)$ Area $= 10 - 2$ Area $= 8$

43 [Challenging] • Target: Curve sketching • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.

a. 1

b. 3

c. **4** ✓

d. -3

The x-intercepts of the graph of $y = f(x)$ are the roots of the equation $f(x) = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$, $b = -4$, and $c = 3$. The sum of the x-coordinates of the intersection points is $-\frac{-4}{1} = 4$.

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44 [Challenging] • Target: Optimization problems • Cross-Lesson: 3.01 The Chain Rule

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.

a. **5** ✓

b. -5

c. 6

d. -6

The x-coordinates of points A and B are the roots of the quadratic equation $f(x) = 0$, which is $x^2 - 5x + 6 = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this equation, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

45 [Challenging] • Target: Optimization: minimizing cost or distance • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?

a. 1

b. 2

c. **3** ✓

d. 4

To find the x-intercepts, we set $f(x) = 0$. This gives the quadratic equation $-x^2 + 4x - 3 = 0$. Multiplying by -1, we get $x^2 - 4x + 3 = 0$. Factoring this equation, we have $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. The larger x-value is 3.

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